

Week 00. Prerequisites for part on quantitative research methods

Part 1: Statistics and Econometrics. MK68-2024/25

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Week 1 Prerequisites

Key prerequisite for this week is the undergraduate level of knowledge in statistics. This includes, but not limited to understanding the following concepts:

- Mean, Median, Mode, Maximum, Minimum, Quantiles, Quartiles, Percentiles;
- Box plot and Histogram;
- Distribution, skew and kurtosis;
- Power ($x^a = b$) and the logarithm transformation $\log_a b = x$. Properties of the logarithm: $\log_a 1 = 0$ and $\log_a 0 = -\infty$.

To bridge any of the knowledge gaps, you may follow materials prerecorded for you and available on Ilias:

1. [Introduction to statistics and scale levels](#). [Slides](#) and [Exercises](#)
2. [Basic measures](#). [Slides](#)
3. [Understanding and visualizing data I](#). [Slides](#) and [Exercises](#)
4. [Understanding and visualizing data II](#). [Slides](#) and [Exercises](#)

Optional materials

If materials mentioned above are insufficient, consider following the unit [Summarizing quantitative data](#) and [Displaying and comparing quantitative data](#) from the [Khan academy](#). Specifically, you are encourage to pay close attention to:

- [Measuring center in quantitative data](#)
- [Interquartile range \(IQR\)](#)
- [Box and whisker plots](#)

- [Displaying quantitative data with graphs](#) with specific focus on the videos:
 - “Creating a histogram”, “Histograms” and “Interpreting a histogram”.
- [Introduction to Logarithm](#)
 - See also [Unit: Exponential & logarithmic functions](#)
- Finally, a wide range of text books is available in the library.

Week 2 Prerequisites

- Mean, Variance, Standard deviation;
- Probability;
- Normal Distribution and confidence intervals;

Pre-recorded videos available on Ilias:

5. [Probability](#). [Slides](#) and [Exercises](#).
6. [Statistical distributions](#). [Slides](#) and [Exercises](#).
7. [Confidence intervals](#). [Slides](#) and [Exercises](#).

Optional materials

- [Variance and standard deviation of a population](#) (29 min) and [a sample](#) (20 min)
- [More on standard deviation](#) (15 min)
- [Basic theoretical probability](#) (30 min)

Week 3 Only optional materials

There no specific materials prerecorded on Ilias. However, to enhance learning experience, one may consider following Khan Academy module on [Modeling data distributions](#) and specifically such topics as:

- [Percentiles](#) (10 min)
- [Z-scores](#) (11 min)
- [Normal distributions and the empirical rule](#) (20 min)